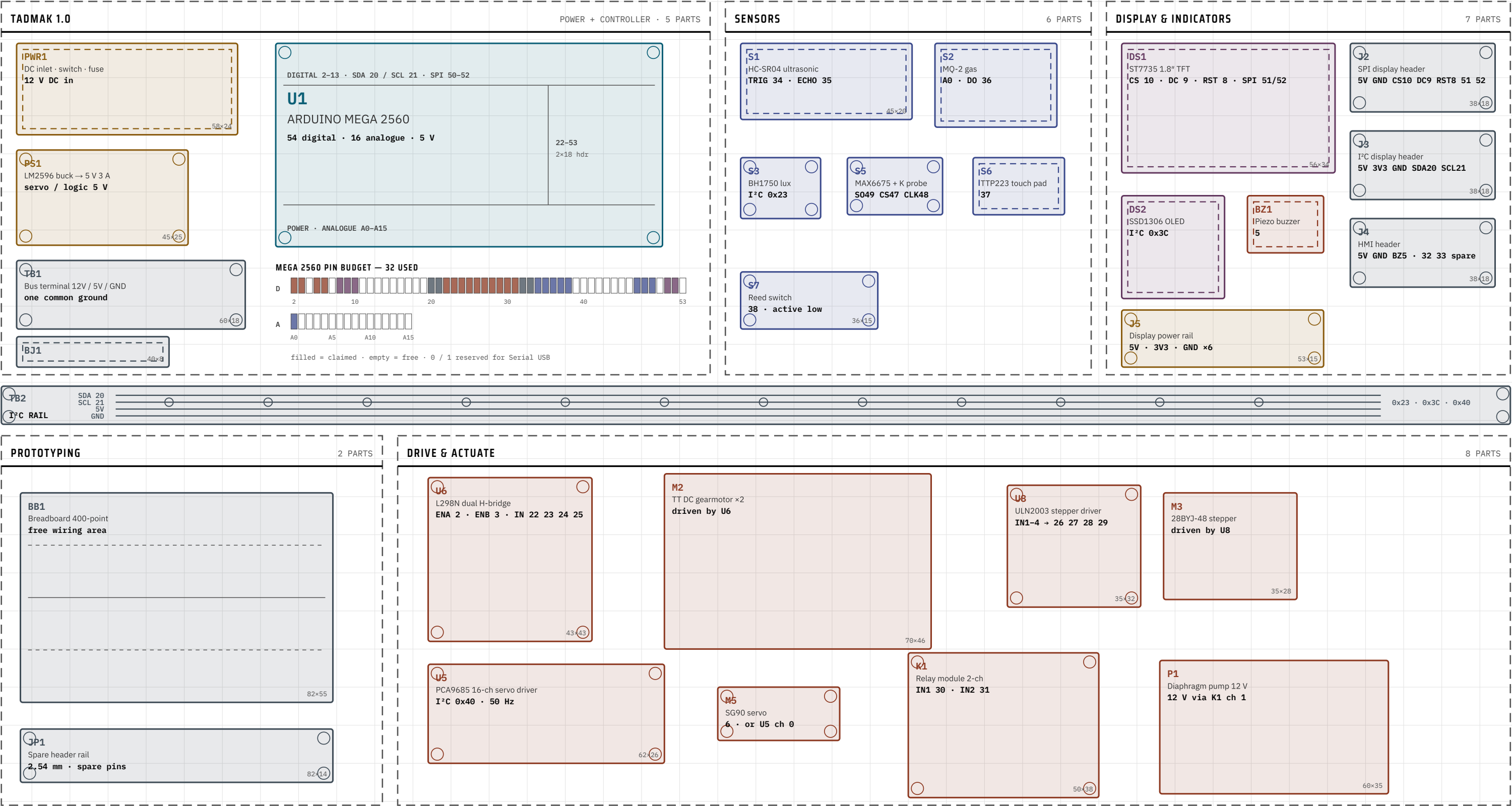


TADMAK 1.0 · ROBOTICS LEARNING KIT FRONT PANEL

ONE MEGA 2560 · PINS SHOWN ON EACH PART



PROTOTYPING

2 PARTS

BB1

Breadboard 400-point

free wiring area

82x55

JP1

Spare header rail

2, 54 mm · spare pins

82x14

DRIVE & ACTUATE

8 PARTS

U6

L298N dual H-bridge

ENA 2 · ENB 3 · IN 22 23 24 25

43x43

M2

TT DC gearmotor x2

driven by U6

70x46

U8

ULN2003 stepper driver

IN1-4 → 26 27 28 29

35x32

M3

28BYJ-48 stepper

driven by U8

35x28

U5

PCA9685 16-ch servo driver

I²C 0x40 · 50 Hz

62x26

M5

SG90 servo

6 · or U5 ch 0

K1

Relay module 2-ch

IN1 30 · IN2 31

50x38

P1

Diaphragm pump 12 V

12 V via K1 ch 1

60x35

NOTES

- TADMAK 1.0 is the brain block: the 12 V inlet, the 5 V supply and the one controller. Everything else on the panel wires back to it at 5 V — no level shifting anywhere.
- Pin numbers printed on each footprint are Mega digital pins. 0 and 1 stay free for Serial over USB.
- Solid outline = module on M3 × 6 standoffs, drill ø3.2 at the four marks. Dashed window = cut through the panel.
- I²C rail runs SDA 20 / SCL 21 across the panel: BH1750 0x23, SSD1306 0x3C, PCA9685 0x40 — no address collisions.
- P1 runs on raw 12 V taken ahead of PS1 and switched by K1 — only the relay's control side touches the Mega. U6, U8 and the servos take 5 V from PS1, never the Mega's 5 V pin. Grounds commoned once, at TB1.
- Pins 44–46 left free: the Servo library claims Timer5 and disables analogWrite there.



TADMAK 1.0 — FRONT PANEL SKETCH

DRAWING No		REV / DATE
DIB-KIT-2026-FP-01		D · 2026-09-22
CONTROLLER	SCALE	SHEET
MEGA 2560 (sole)	1 : 1	ISO A3 · mm
DRAWN	REPO	
Jatin · M. Arora	2026/layout	